

Human Pose Estimation & Correction During Exercise Using ML

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Abstract- The fusion of machine learning and computer vision has enabled sophisticated methods for monitoring and enhancing human movement. This study investigates the use of real-time pose estimation to support alignment correction during physical activities. Through standard camera input and pose recognition algorithms, key anatomical landmarks are detected and evaluated for accuracy against predefined movement patterns. A similarity-based approach, such as cosine comparison, is employed to gauge pose correctness, with a live timer triggered upon successful alignment to measure consistency over time. An interactive graphical interface supplements the system by offering real-time visual feedback, promoting user engagement and self-guided correction. In parallel, the paper explores the evolution of human pose estimation, a growing domain with broad applications in health, sports, and interactive systems. By harnessing deep learning models, the proposed solution introduces an AI-driven trainer capable of detecting posture deviations and delivering corrective feedback dynamically. This integrative framework aims to enhance user performance, encourage proper form, and provide a foundation for intelligent, personalized physical training experiences.

Keywords – Pose Estimation, Computer Vision, Cosine Similarity, Keypoint Detection, Machine Learning , Suggestion System.

I. INTRODUCTION

This project introduces an innovative, AI-driven virtual training assistant that revolutionizes home workouts and rehabilitation by providing real-time feedback on posture through a standard webcam interface.

Developed with a Flask back-end and a Node.js front-end, the application is meticulously designed to detect and rectify users' exercise form utilizing sophisticated human pose estimation and neural network-based evaluation. At its core, the system harnesses the Move Net Thunder model for precise body key-point tracking, capturing critical landmarks from the user's live video feed. When a user selects one of the seven available exercises and initiates movement, the system activates real-time camera access, extracts pose vectors, and juxtaposes them against trained reference data via a deep learning pipeline. The model, engineered with ReLU activation and dropout regularization for enhanced robustness, assesses the congruence of the user's pose through feature vectors. Upon identifying a correct alignment, the pose feedback module visually affirms success—typically by illuminating the overlay in green—and commences a timer to monitor pose-holding duration,

thereby fostering both precision and consistency in form. In addition to pose correction, the system integrates a pain-responsive recommendation feature. Users can specify a particular body region experiencing discomfort (e.g., shoulder, lower back), and the system astutely returns a curated list of exercises tailored to alleviate that specific pain. This functionality is underpinned by a machine-learned suggestion engine trained on therapeutic pose-to-pain mappings, rendering the solution invaluable not only for fitness enthusiasts but also for physiotherapy and recovery support.

The platform boasts a sleek, user-friendly interface and amalgamates real-time computer vision with actionable feedback, making it an ideal resource for individuals lacking access to personal trainers. Whether for novices striving for proper form, patients requiring guided rehabilitation, or fitness aficionados aiming to refine their technique, this system functions as a virtual mentor—enhancing performance, mitigating injury risk, and personalizing the exercise experience through intelligent automation.

II. LITERATURE SURVEY

1. The objective of the paper "Machine Learning-Based Exercise Posture Recognition System Using MediaPipe Pose Estimation Framework," published in IEEE on March 18, 2023, by Weeriya Supanich and Suwanee Kulkarineetham, was to devise a posture classification system that meticulously analyzes exercise movements utilizing video input captured from a webcam. The system employs MediaPipe to extract skeletal keypoints, which are subsequently processed through various machine learning models for classification. The efficacy of these models was assessed using performance metrics such as precision, recall, and accuracy. A notable advantage of the study is its capacity to deliver automated feedback, thereby eliminating the necessity for a personal trainer. However, the research is not without its limitations. Its dependence on static datasets for posture labeling.
2. The aim of the paper "Enhancing 3D Pose Estimation Accuracy from Multiple Camera Perspectives through Machine Learning Model Integration", published in IEEE on 29 April 2023 by Ervinas Gislis and Arturas Serackis, was to improve 3D human pose estimation by integrating data from two cameras placed at orthogonal angles. The authors utilized a three-layer feed-forward neural network to refine joint coordinate predictions by merging outputs from both camera perspectives. This integration significantly reduced estimation errors, enhancing overall pose detection accuracy. However, the system's effectiveness is dependent on proper multi-camera calibration, limiting its scalability for consumer-level applications.
3. The objective of the paper "Pose Trainer: Correcting Exercise Posture using Pose Estimation", published in IJCRT on 20 April 2021 by Stivard Chin and Richard Yang, was to develop an application that provides real-time feedback on exercise posture. The system detects body posture using pose estimation and evaluates it through geometric vector analysis to guide the user in correcting form. A dataset containing over 100 videos with correct and incorrect exercise forms was recorded for training. While the system offers accurate feedback on four exercises and is compatible with standard hardware, its main limitation is the narrow exercise range and lack of mobile compatibility.
4. The aim of the paper "AI Fitness Trainer Using Human Pose Estimation", published in IJERT on 13 December 2023 by Abhinand G, Mohammed Anas, and Naveen Kumar, was to design a real-time fitness guidance system that uses MediaPipe and BlazePose to evaluate human posture during exercise routines. The system provides immediate corrective feedback, enhancing safety and effectiveness for at-home workouts. Its primary benefit is reducing the reliance on physical trainers, but the limitation lies in the need for continuous updates to accommodate a wide variety of exercises and user body types.
5. The objective of the paper "Introduction to Camera Pose Estimation with Deep Learning", published in IJERT on 13 May 2023 by Yoli Shavit and Ron Ferens, was to survey modern deep learning-based methods for pose estimation. The paper compares several state-of-the-art models and outlines advancements made in pose regression techniques. It provides a thorough cross-comparison of existing frameworks and highlights key trends in improving pose accuracy. The main limitation discussed is the generalization challenge when models are applied across diverse environments and movement patterns.
6. The aim of the paper "LogRF: An Approach to Human Pose Estimation Using Skeleton Landmarks for Physiotherapy Fitness Exercise Correction", published in IEEE Access in 2023 by Ali Raza et al., was to introduce a system that utilizes skeletal landmarks to aid in physiotherapy and fitness posture correction. The proposed model, LogRF, supports the rehabilitation process by analyzing deviations in human pose and providing feedback for posture correction. While the method offers great promise for therapy applications, it requires fine-tuning for broader applicability across multiple demographic groups.

III. PROBLEM STATEMENT

Maintaining correct posture during physical exercises is crucial for effectiveness, injury prevention, and long-term physical health. However, without access to expert supervision, individuals often perform exercises with improper form, which can lead to injuries or reduce exercise efficiency. Existing solutions like fitness apps and instructional videos lack real-time feedback or personalization, making it difficult for users to self-correct during activity. There is a need for an intelligent system capable of detecting human body posture in real

time, analyzing deviations from ideal poses, and providing immediate corrective feedback, all within an accessible, user-friendly platform.

displays feedback dynamically. The result is a robust, interactive system.

IV. PROPOSED WORK

The proposed system aims to deliver an intelligent, real-time exercise posture correction and recommendation platform that enhances personal fitness training and rehabilitation through computer vision and deep learning. The core of the system revolves around the use of **MoveNet Thunder/Lightning pose estimation models** integrated with a **custom-trained neural network** to detect, evaluate, and provide feedback on human body posture during physical exercises. Upon selecting any of the seven predefined exercises from the interface, the system activates the user's webcam to capture live video. Using pose detection algorithms, it extracts **body keypoints (landmarks)** and compares them with a reference dataset using **feature vector mapping** and **cosine similarity** for accurate pose matching. The evaluation model includes **ReLU-activated dense layers** and **dropout** to reduce overfitting and enhance generalization across different users.

When the user's posture aligns closely with the correct form, the system highlights the pose in green and starts a **real-time counter** to track how long the correct posture is maintained—helping measure both **accuracy and endurance**. Incorrect postures trigger visual alerts, allowing the user to self-correct immediately.

Additionally, the platform introduces a **pain-based recommendation engine**. Users can input pain-related terms such as "lower back pain" or "shoulder stiffness", and the system suggests suitable exercises tailored to relieve discomfort in the specified area. This is achieved using a simple **rule-based mapping algorithm** integrated with the exercise database.

The entire pipeline is powered by a **Flask backend**, responsible for model inference and data handling, and a **Node.js frontend**, which handles UI interactions and

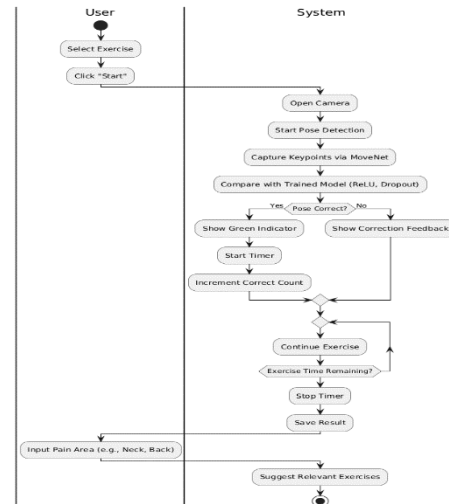


Fig-1: SYSTEM FLOW

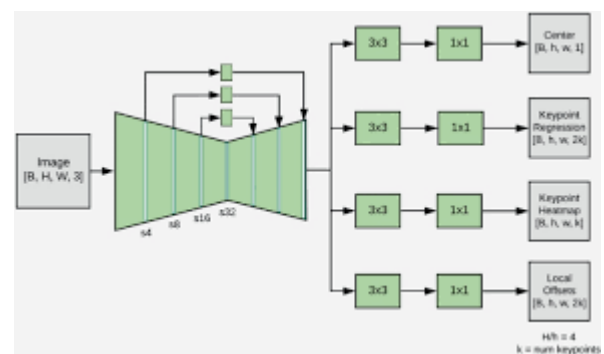


Figure2: POSE ESTIMATION AND LAYER VIEW

1. REQUIREMENT ANALYSIS

- ◆ Software Requirements:
 - Python 3.7 and above
 - Flask Framework
 - Node.js (for frontend)
 - TensorFlow / TensorFlow.js
 - OpenCV
 - MediaPipe/MoveNetThunder or Lightning model
 - Web Browser (Chrome/Edge/Firefox)
 - OS: Windows 7/8/10, Linux, or macOS
- ◆ Hardware Requirements:
 - Minimum 8 GB RAM
 - Processor Speed: 2.0 GHz and above

- Integrated or Dedicated Webcam
- GPU (optional for faster model training/testing)

◆ **Technology Used:**

- Python (Flask for backend API)
- Node.js /JavaScript (for frontend development)
- HTML/CSS (UI design)
- TensorFlow & MediaPipe (pose estimation)
- ReLU Neural Network and Cosine Similarity (pose comparison).
- React.js (optional, for SPA frontend).
- MachineLearning for pain-based recommendation system .

RESEARCH AND METHODOLOGY

The research is centered around developing an intelligent, AI-driven web application for human pose estimation and correction during physical exercises. The system is designed to provide real-time feedback using computer vision and machine learning techniques integrated into a Flask–Node.js architecture.

A. Dataset Collection and Preprocessing:

A curated dataset containing annotated poses for seven exercises is used as reference. The dataset is preprocessed to extract body keypoints using MediaPipe and MoveNet Thunder/Lightning models. These keypoints are normalized and vectorized to form a consistent training input.

B. Pose Detection and Feature Extraction:

Live video feed is captured via webcam, and keypoints are extracted using the MoveNet model. These keypoints represent various joint positions of the body, which are used to form feature vectors for pose comparison.

C. Pose Comparison Algorithm:

A ReLU-activated neural network processes the extracted features and compares them to reference poses using cosine similarity. This determines the closeness of the user's pose to the correct one. When the similarity threshold is met, the system highlights the pose (e.g., turning green) and initiates a time counter to evaluate pose-holding duration.

D. Feedback Mechanism:

The frontend interface, built using Node.js and optionally React.js, provides immediate visual and textual feedback to the user. Incorrect posture triggers alerts and guidance

for improvement, while correct posture is confirmed and tracked for consistency.

E. Pain-Based Recommendation System:

An additional feature allows users to input specific body pain areas (e.g., back, shoulder, knee). The system uses a machine learning model trained on labeled data to recommend appropriate exercises known to alleviate the specified discomfort.

F. Model Training and Optimization:

The neural network is trained using a dropout strategy to prevent overfitting, and the train-test split ensures balanced performance evaluation. Performance metrics such as accuracy, precision, and recall are used to fine-tune model parameters.

V. RELEVANT MATHEMATICS ASSOCIATED WITH THE PROJECT

1. Linear Algebra

Tensors and Matrices: Pose keypoints are captured as multidimensional arrays (tensors). Operations such as matrix multiplication are essential during neural network computations.

Flattening: Conversion of 2D keypoint coordinates (x, y) into a 1D feature vector is done by reshaping tensors for input into the neural network.

Normalization: Keypoint coordinates are scaled to a fixed range, typically [0, 1], using the normalization formula:

$$x_{norm} = \frac{x - \min}{\max - \min}$$

2. Geometry and Trigonometry

Pose Angles: The angle between three joints (e.g., hip–knee–ankle) is calculated

using the cosine of the angle between vectors:

$$\cos(\theta) = \frac{\vec{A} \cdot \vec{B}}{\|\vec{A}\| \|\vec{B}\|}$$

Distance Calculation: The Euclidean distance between two keypoints is computed

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

3. Vector Mathematics

Feature Vectors: Each pose is represented as a concatenated vector of all 2D key-points, enabling use in classification models.

Cosine Similarity: To compare similarity between user and reference poses:

$$\text{Similarity} = \frac{A \cdot B}{\|A\| \|B\|}$$

4. Neural Network Mathematics

Feedforward Propagation: The output at each layer is computed as:

$$Z = W \cdot X + b, \quad A = f(Z)$$

Activation Function (ReLU):

$$\text{ReLU}(x) = \max(0, x)$$

Dropout Regularization: Randomly disables a fraction of neurons during training to reduce overfitting.

5. Probability & Classification

Softmax Function: Converts the final outputs (logits) into probability distributions:

$$\text{Softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

Cross-Entropy Loss: Measures the difference between predicted and actual class probabilities.

6. Optimization Algorithms

Gradient Descent (Backpropagation): Used to update neural network weights by computing derivatives of the loss function.

7. Time-Series & Performance Metrics

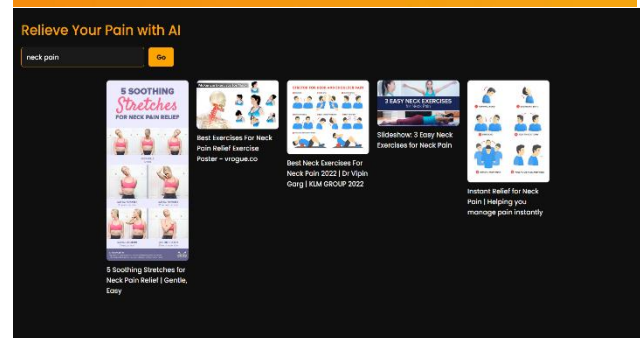
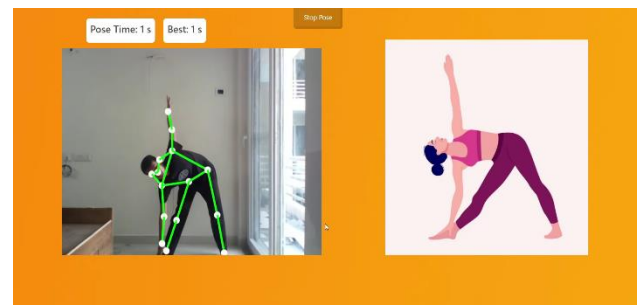
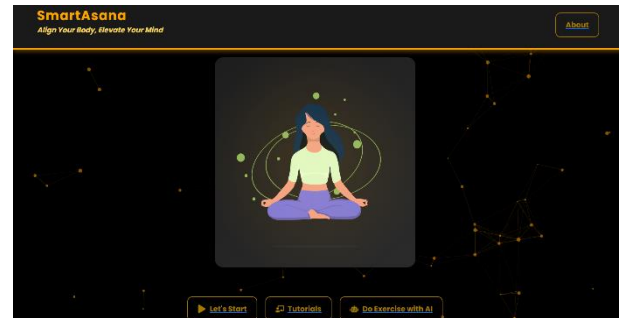
Correct Pose Duration: Simple arithmetic and condition tracking are used to mea-

sure the duration for which the user holds a correct pose.

Model Evaluation Metrics: Accuracy, precision, recall, and F1-score are computed

to assess the classification model's performance.

VI. RESULTS



FUTURE WORK

Further improvements can include expanding the range of supported exercises and integrating more advanced pose estimation models for higher precision. Incorporating personalized fitness plans based on user history and adaptive learning algorithms can enhance the system's responsiveness over time. Adding voice-guided feedback and support for multi-user sessions may increase usability and engagement. Additionally, integrating wearable sensor data and enabling mobile application deployment could make the system more versatile and accessible for diverse user needs, including elderly care and post-injury rehabilitation.

VII. CONCLUSION

An intelligent system for real-time human pose estimation and correction has been developed using a

Flask backend and Node.js frontend. Leveraging the MoveNet Thunder model for precise keypoint detection and a ReLU-activated neural network for posture classification, the platform offers accurate visual feedback during seven predefined exercises. Correct postures are indicated with color changes and time tracking, enhancing user interaction and monitoring consistency. Additionally, a pain-based recommendation module suggests targeted exercises based on user-inputted discomfort areas. The integration of dropout techniques and thorough train-test validation supports system robustness and adaptability, making it suitable for fitness, therapy, and guided home workout

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REFERENCES

1. Weeriya Supanich and Suwanee Kulkarineetham, "Machine Learning-Based Exercise Posture Recognition System Using MediaPipe Pose Estimation Framework," *IEEE*, March 18, 2023.
2. Ervinas Gisleris and Arturas Serackis, "Enhancing 3D Pose Estimation Accuracy from Multiple Camera Perspectives through Machine Learning Model Integration," *IEEE*, April 29, 2023.
3. Stivard Chin and Richard Yang, "Pose Trainer: Correcting Exercise Posture using Pose Estimation," *IJCRT*, April 20, 2021.
4. Abhinand G, Mohammed Anas, and Naveen Kumar, "AI Fitness Trainer Using Human Pose Estimation," *IJERT*, December 13, 2023.
5. Yoli Shavit and Ron Ferens, "Introduction to Camera Pose Estimation with Deep Learning," *IJERT*, May 13, 2023.
6. Ali Raza, Azam Mehmood Qadri, Iqra Akhtar, Nagwan Abdel Samee, and Maali Alabdulhafith, "LogRF: An Approach to Human Pose Estimation Using Skeleton Landmarks for Physiotherapy Fitness Exercise Correction," *IEEE Access*, 2023.
7. Xiao Chu, Wanli Ouyang, Hongsheng Li, Xiaogang Wang, "Structured Feature Learning for Pose Estimation," *IEEE Access*, 2016.
8. Jianbo Wang, Kai Qiu, Houwen Peng, Jianlong Fu, Jianke Zhu, "AI Coach: Deep Human Pose Estimation and Analysis for Personalized Athletic Training Assistance," *MM*, 2019.
9. Federico Roggio, Bruno Trovato, Martina Sortino, "Comprehensive Analysis of the Machine Learning Pose Estimation Models Used in Human Movement and Posture Analyses: A Narrative Review," *Heliyon*, 2024.
10. Rutuja Gajbhiye, Snehal Jarag, Pooja Gaikwad, Shweta Koparde, "AI Human Pose Estimation: Yoga-aasan Pose Detection and Correction," *IJISRT*, 2022.
11. Zhehan Kan, Shuoshuo Chen, Ce Zhang, Yushun Tang, Zhihai He, "Self-Correctable and Adaptable Inference for Generalizable Human Pose Estimation," *arXiv*, 2023.
12. Esraa Samkari, Muhammad Arif, Manal Alghamdi, Mohammed A. Al Ghamdi, "Human Pose Estimation Using Deep Learning," *MDPI*, 2023.
13. Alexander Toshev, Christian Szegedy, "DeepPose: Human Pose Estimation via Deep Neural Networks," *CVPR*, 2014.
14. Linzhi Huang, Yulong Li, "Semi-Supervised 2D Human Pose Estimation Driven by Position Inconsistency Pseudo Label Correction Module," *arXiv*, 2023.
15. Leonid Sigal, "Human Pose Estimation," *Disney Research*, Pittsburgh.
16. Santosh Kumar Yadav, Amitojdeep Singh, Abhishek Gupta, "Real-time Yogaasan Recognition Using Deep Learning," *Springer*, 2019.
17. Umer Rafi, Ilya Kostrikov, Juergen Gall, "An Efficient Convolutional Network for Human Pose Estimation," 2016.

18. Muhammad Usama Islam, Hasan Mahmud, Faisal Bin Ashraf, Iqbal Hossain, Md. Kamrul Hasan, "Posture Recognition by Detecting Human Joint Points in Real Time Using Microsoft Kinect," 2020.
19. Shih-En Wei, Varun Ramakrishna, Takeo Kanade, Yaser Sheikh, "Convolutional Pose Machines," CVPR, 2016.
20. Yang Liu, Changzhen Qiu, Zhiyong Zhang, "Deep Learning for 3D Human Pose Estimation and Mesh Recovery," arXiv, 2024.
21. Rahul Ravikant Kanase, Akash Narayan Kumavat, Rohit Datta Sinalkar, Sakshi Somani, "Pose Estimation and Correcting Exercise Posture," ITM, 2021.
22. Wenyang Hu, Kai Liu, Libin Liu, Huiliang Shang, "Spatial-Temporal Transformer-based Framework for Human Pose Assessment and Correction in Education Scenarios," arXiv, 2023.
23. Vikas Gupta, "Back to the Future: Joint Aware Temporal Deep Learning 3D Human Pose Estimation," 2022.
24. Eric Hedlin, Helge Rhodin, Kwang Moo Yi, "A Simple Method to Boost Human Pose Estimation Accuracy by Correcting the Joint Regressor for the Human Dataset," 2021.
25. Jonathan Tompson, Arjun Jain, Yann LeCun, "Joint Training of a Convolutional Network and a Graphical Model for Human Pose Estimation," arXiv, 2014.
26. Alejandro Newell, Jia Deng, "Pixels to Graphs by Associative Embedding," arXiv, 2018.
27. Jamie Shotton, Andrew Fitzgibbon, Mat Cook, Toby Sharp, Mark Finocchio, "Real-Time Human Pose Recognition in Parts from Single Depth Images," Xbox Inc., 2011.
28. Federica Bogo, Angjoo Kanazawa, Christoph Lassner, Peter Gehler, Javier Romero, Michael J. Black, "Keep it SMPL: Automatic Estimation of 3D Human Pose and Shape from a Single Image," arXiv, 2016.
29. Jessika, Astri Handayani, Isca Amanda, Hasnaop Auliya, "A Study on Part Affinity Fields Implementation for Human Pose Estimation with Deep Neural Network," 2020.
30. George Papandreou, Tyler Zhu, Liang-Chieh Chen, Spyros Gidaris, Jonathan Tompson, Kevin Murphy, "PersonLab: Person Pose Estimation and Instance Segmentation with a Bottom-Up, Part-Based, Geometric Embedding Model," ECCV, 2018.
31. Petrisa Zell, Bastian Wandt, Bodo Rosenhahn, "Joint 3D Human Motion Capture and Physical Analysis from Monocular Videos," CVPR, 2018.
32. Zhe Cao, Gines Hidalgo, Tomas Simon, Shih-En Wei, Yaser Sheikh, "OpenPose: Realtime Multi-Person 2D Pose Estimation Using Part Affinity Fields," arXiv, 2019.